EXOSKELETON

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Abstract— as today's technology is developing day by day we could see a drastic change in the world's revolution in technology. Whereas day by day the technology is replacing with different and advanced features the people could experience. Therefore "EXOSKELETON" which is well known and successful in the health care industry, which is a haptic exoskeleton that is used to make paralyzed patients walk again. This exoskeleton system is dominating the healthcare industry with its advanced and durable features and application which is successful. This clearly gives and outcome in humans as the technology interacts directly with the humans. Therefore the Introduction, History, Structure, Future and the references are clearly mention with examples of images to make it more clear about the Exoskeleton technology.(Abstract)

Keywords— Exoskeleton, Paralyzed, applications, Dominating, Replacing, Features, Technology (*key words****)***

I. INTRODUCTION

The Exoskeleton is design for the purpose to make the paralyzed patients walk. This technology is specially designed and found by the medical industry to help the people who suffer from the disability of not walking back and this technology has the specialty to support them in walk back again with a support of a machine. This technology is been used over a decade to help the patients who are in need of help and seeking a support to walk back again. This is the kind of robotic structure which will be attached to the joints to make the muscles power up as it is needed. This supports the back, shoulders, waist and thigh support. And also assists the movement of the patient which is known as the user motion. This can be the movements like lifting , moving forward, moving backward, sitting, walking in the ways that the person wants and able to move the body movement. [1]

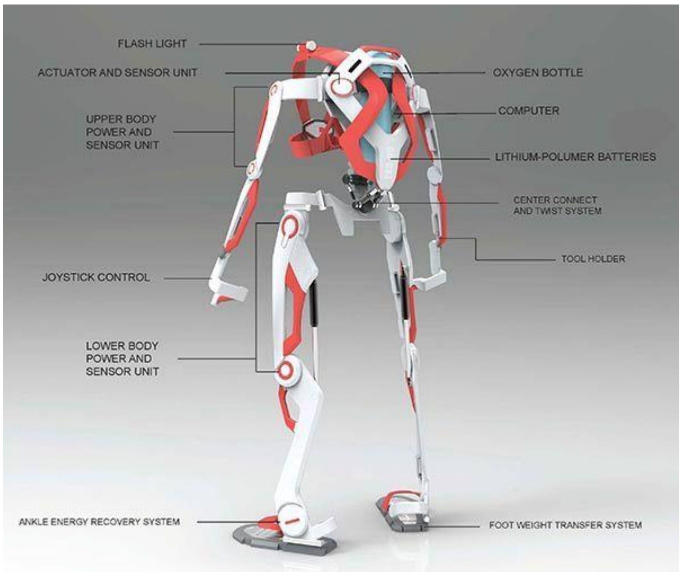
This basically attached to the joints of the body which supports the powerless bones to power and make the muscle active and make a movement of the body. This technology has changed many lives and this has been a very successful machine in the medical industry. After many tests and examination this was successful on patients to be used. After all the technologies being found this one is a life turning point for the people who think that they can never walk back again. [1]

Figure 2: This image shows the exoskeleton be used on a patient to keep him walk, [1]

II. HISTORY OF EXOSKELETON

This Exoskeleton technology was found in 2005 by a California based company called EKSO BIONICS. Whereas the doctors say that this company has made it possible for the patients who wants to walk back again, if not this would not have been possible to make it happen. This is a life changing movement for the people who are in seek of help to walk back after losing their life in some incident and never walk back again. And this company has made it happen for the needs. This Exoskeleton technology is designed for the each part of the body in any motion of movement that the part of body wants to make.

"Miraculous help for the paralyzed" says the doctors who made the patients try this and the outcome was so fantastic. [1]



III. STRUCTURE

The structure of the Exoskeleton is designed in the form of capturing the movements made by the person using it which

Figure 1: Structure of an Exoskeleton, [4]

is also known as task oriented whereas the movements will be improved as the performance of patient also improves. This is witnessed according to the neurological lesions. [2]

The structure and the application of the robotics can be assisted and evaluated according to the movement which will be captured by the used forced sensors. These sensors which are implemented in this technology will sensor each movement and are optimized based on the architecture. [2]

The above placed image shows the structure how and what it is made of and looks like. This structure is designed according to the human body structure, inured to make the patients comfortable using this and not making it more difficult for them. This structure will just look same as a robotic. Whereas this is a robotic technology which will be attached to the human body to make movements and sensor the motions.

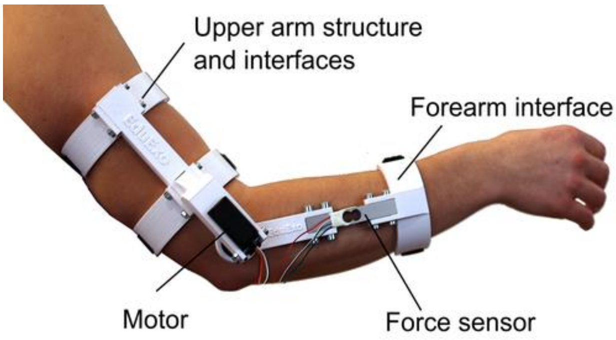


Figure 3: Exoskeleton for arm, [5]

The figure 3 shows the image of an arm where the exoskeleton is been used. And also it shows the upper arm interface and the forearm interface in order to give the arm a support for the movement and the force sensors are placed to sensor the movement according to movement of motion made by the arm. The motor is placed to make the arm work and move, this is because the exoskeleton needs a way to power up so this is way which it can be made.

Exoskeleton is also known as exoskeletal robots which are wearable. This kind of mechanical structure is conceptually made into a skeletal structure according to the human body. Which includes the limb and the other involved body parts. [3]

This Exoskeleton robots wear found and designed to closely interact with the human's in order to fulfill the needs and the purpose to assist with the day to day activities in performance. These exoskeleton robots are assistive which can be used permanently in carrying out the most important activities in day to day living. This is for small children and elders who are paralyzed. [3]

IV. Future

As the technology improves day by day the structure and the application of exoskeleton is been changing and making it simpler as per the usage of the technology. It is said that the current situation of healthcare is based on the technology because, the technology is used in every form in the health care system such radiology , MRI, X-Rays, ECO,etc. all of these are designed for the purpose of the medical treatment in the healthcare industry. Therefore technology plays a major role in the healthcare industry.

According to the current situation the technology is ruling the world even in day to day living of life. As per the healthcare industry this exoskeleton can change and vary accordingly to the person's need. And this can be made even more advanced with all the equipment that would make even more efficient.

Efficiency is the most important part in this exoskeleton , because if the machine does not work efficient then the person will not be able to make progress in the movements and improve the performance of his/her stability. Exoskeleton is the way that creates the ability for the paralyzed person to make movement front and back.

In future the design and application of the exoskeleton could be changed as per the advanced technology. Such as advanced sensors and motors to make it even more efficient. This can lead to a productive and efficient service provide in healthcare industry. Hence this is a life changing process for the paralyzed of their hope becoming true to walk back again. In future the technology can even be simpler where the size, appearance, structure, applications and many other used to build the exoskeleton can be replaced with advanced features and made even more comfortable, efficient, productive and successful.

This Exoskeleton also could be manufactured/ made in different form, like separately for each body part in specific to the patients paralyzed body part. Which can be utilized according to the patient body structure and made it comfortable for them.

It can be for accessible as per the technology components used in the skeletal structure. Whereas this will lead the future to an advanced healthcare system.

Figure 4: The future of Exoskeleton, [7]

The more advanced the technology is the simpler the life becomes. In this picture you could see the development and the advancement of the technology accordingly. The technology changes from a wheelchair and advances to a exoskeleton structure which made the paralyzed patient walk again. So they will not have spend their entire life in wheelchair unable to enjoy stuffs they like and not able to live a normal life. But the advancement of technology is made it easier whereas the paralyzed patient do not have to worry about spending the whole life in wheelchair instead they could wall again with the help of exoskeleton.

Figure 5: Medical Exoskeleton, [8]

This images shows the structure of a medical Exoskeleton, how it is used in the medical industry. This medical Exoskeleton will be weared in the legs of the paralyzed patient and they will be given test to walk with the Exoskeleton with instructions and will be examined by the doctors accordingly where the movements of the body will be captured.

V. References

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